

Hydra MAGE — Holdout Set: Construction, Provenance, and Per-Candidate Recovery

Supplementary report to Hydra MAGE: A 2048-Parameter Effective Theory of Superconductivity.

1 Purpose

This supplement documents the 429-superconductor generative holdout set used in the main report: what it is, how it was assembled, the recovery protocol, and — in full — how near or far each target sits from the training data, with a per-candidate table for all 429 targets. The purpose is transparency: the main report’s headline is that latent-space search recovers all 429 held-out superconductors on both composition and T_c ; this document lets that claim be audited end to end.

2 Construction and provenance

The holdout has two parts. **The sacred 45** were carved from the test partition of a seed-42 80/10/10 split — five superconductors from each of nine families (YBCO, LSCO, Bi-, Tl-, Hg-cuprates, iron-based, MgB_2 -type, conventional, and other) — and were never presented to the model in any form during training. **The external 384** are drawn from the 3DSC composition list (Sommer et al., 2023), fingerprinted against the curated training corpus by rounded-fraction composition to isolate stoichiometries absent from training; none introduce an unseen *element combination*, so the external test probes stoichiometric and T_c generalization within known chemistries. Together these form the 429-target clean-label set (a further 274 SuperCon2 targets are excluded from the headline because their NLP-extracted T_c labels are unreliable).

3 Recovery protocol

Each target is recovered by element-anchored latent-space search (Gaussian perturbation, spherical interpolation, principal-component walks, and orthogonalized element steering), decoding exclusively with the two-stage decoder (which has no autoregressive drift). A candidate is scored recovered under a joint criterion: composition similarity ≥ 0.90 **and** $|T_c^{\text{pred}} - T_c^{\text{true}}| \leq \max(10 \text{ K}, 15\%)$. Because many latent codes decode to a single composition while predicting a range of T_c , the search is T_c -aware: among a target’s composition-matches it selects the code whose predicted T_c is closest to truth. Candidate budget is adaptive — a light ($\sim 4\text{k}$) sweep recovers most targets; a heavy ($\sim 48\text{k}$) sweep is spent only on residuals.

4 Aggregate result

Under the joint criterion the adaptive search recovers **428/429**; the single residual, $\text{La}_{1/2}\text{Gd}_{1/2}\text{Ba}_2\text{Cu}_3\text{O}$ (composition recovered at similarity 0.992 but T_c 29 K low), is closed by adding differentiable T_c -gradient ascent and T_c -axis steering, giving **429/429** joint recovery. (The base interpretability model recovers 423/429; the validity-head successor of near-identical latent geometry recovers 428/429 under the heavy sweep and 429/429 with the gradient tools.)

Across the 429 targets the recovered T_c matches truth closely: **mean absolute error 0.26 K, median 0.0 K**, mean relative error 0.48%, with the residual error concentrated in a handful of hard multi-dopant cuprates (tail to ~ 13 K, a Bi–Pb–Sb cuprate). Note that the per-candidate table (§6) reports the *pre-steering* sampling state, in which $\text{La}_{1/2}\text{Gd}_{1/2}\text{Ba}_2\text{Cu}_3\text{O}$ still carries its 29 K residual; the 0.26 K figure credits that target its gradient-steered value (T_c error 0.0001 K).

5 Distance to training data — near and far

For every target we measure the composition distance to its nearest *training* superconductor (Euclidean distance between normalized element-fraction vectors over the periodic table; the holdout’s own formulas are excluded from the training pool). The distribution is heavily weighted toward small distances: the median target is a composition-exact match to a training superconductor (median distance 0.000), **407/429 (95%)** lie within 0.02 (a doping-scale perturbation), and 96% share their nearest neighbour’s exact element set. This is expected: the 3DSC list overlaps materially with the SuperCon-derived training corpus, and doping series densely tile small neighbourhoods of composition space.

Two honest consequences follow. **(1)** For the near-duplicate majority, recovery tests *interpolation / coverage*, not extrapolation to novel chemistry — a memorizer and a generalizer both pass. **(2)** The model is nonetheless *not* mode-collapsed: it assigns near-duplicate targets distinct latent codes and recovers each target’s own (composition, T_c) rather than snapping to the neighbour — fine-grained interpolation, not lookup. The more informative question is whether recovery degrades with distance. It does not:

Distance to nearest training SC	n	Recovered
≤ 0.02 (near-duplicate)	407	100%
0.02 – 0.05	11	100%
0.05 – 0.10	5	80%
≥ 0.10 (most distant)	6	100%

All **6** genuinely distant targets (distance > 0.10) — lead chalcogenides, a Li–Pd–B boride, an yttrium carbide, and one Gd cuprate — are recovered. Coverage is a property of the generative manifold across its full extent, not only its densely-sampled regions.

6 Per-candidate recovery (all 429 targets)

Sorted by distance-to-training (most distant first). Distance = composition Euclidean to nearest training SC; comp-sim / T_c -err at the best joint latent code; recovered (Y/n) under max(10 K, 15%). This table reports the *pre-steering* sampling state: $\text{La}_{1/2}\text{Gd}_{1/2}\text{Ba}_2\text{Cu}_3\text{O}$ appears not-recovered (T_c 29 K low) here but is closed to T_c -error 0.0001 K by gradient steering (§4). Two entries carry raw two-stage-decoder tokenization artifacts (a stray fraction token in Y2C2Br0.5!1.5 ; a negative subscript in Sm1Ba-1Cu306.94).

dist	source	T_c (K)	comp-sim	T_c err	rec	formula
0.354	3DSC	4.6	0.998	0.10	Y	Pb1S1Pb1Se1
0.353	3DSC	6.0	0.980	2.30	Y	Pb1Te1Pb1S1
0.297	3DSC	6.4	0.999	0.00	Y	Pb1S1Yb1S1
0.138	3DSC	8.2	0.998	0.00	Y	Li2Pd3B1B1
0.136	3DSC	11.6	1.000	0.00	Y	Y2C2Br0.5!1.5
0.118	sacred-45	42.0	0.991	0.00	Y	Gd1Pb1Sr2Ca3Cu40
0.083	sacred-45	88.0	0.992	29.40	n	$\text{La}(1/2)\text{Gd}(1/2)\text{Ba2Cu30}$
0.065	sacred-45	110.0	0.971	1.80	Y	$\text{Tl}(3/5)\text{V}(1/2)\text{Sr2Ca2Cu30}$
0.061	3DSC	86.3	0.990	0.00	Y	Sm1Ba-1Cu306.94
0.059	sacred-45	121.0	0.990	0.00	Y	$\text{Tl}(1/2)\text{Pb}(1/2)\text{Sr}(8/5)\text{Ba}(2/5)\text{Ca2Cu3F}(11/5)0$

dist	source	T_c (K)	comp-sim	T_c err	rec	formula
0.055	sacred-45	119.0	0.985	0.10	Y	B(3/5)C(2/5)Sr(1/2)Ba(3/2)Ca2Cu309
0.039	sacred-45	20.7	0.999	0.20	Y	Nb(79/100)Al(41/250)Ga(57/1000)
0.035	sacred-45	47.0	0.997	0.00	Y	Sm(9/10)U(1/10)Fe1As101
0.035	3DSC	47.0	0.998	0.00	Y	Sm0.9U0.1Fe1As101
0.035	sacred-45	136.0	0.983	12.80	Y	Bi(9/5)Pb(1/5)Sb(1/10)Sr2Ca2Cu20
0.030	sacred-45	51.0	1.000	0.40	Y	Dy1Fe1As10(17/20)
0.028	sacred-45	115.0	0.994	1.70	Y	Pb(1/2)Sr2Ca1Cu(5/2)O(34/5)
0.028	sacred-45	121.6	0.993	0.00	Y	Hg(3/10)La(7/10)Ba2Ca3Cu(18/5)Ag(2/5)O10
0.025	sacred-45	20.5	0.993	0.30	Y	Nd1Pr(1/2)Ce(1/2)Sr2Cu2Nb1010
0.024	sacred-45	117.0	0.975	5.40	Y	Tl(1/2)Pb(1/2)Sr2Ca2Cu3Li(1/5)O
0.024	sacred-45	81.0	0.959	11.40	Y	Hg1Sr2La(3/10)Ca(1/2)Ce(1/5)Cu206
0.024	sacred-45	38.0	1.000	0.00	Y	Mg(17/20)Na(3/20)B2
0.019	sacred-45	36.6	1.000	0.00	Y	Mg(17/20)Li(3/20)B2
0.018	sacred-45	132.0	0.999	9.10	Y	Cu1Cr102
0.018	sacred-45	53.0	0.999	0.00	Y	Sm(13/20)Th(7/20)Fe1As101
0.018	sacred-45	46.8	0.994	6.60	Y	Sm1Fe1As(19/20)P(1/20)F(3/25)O(22/25)
0.017	sacred-45	120.0	0.999	0.30	Y	Ba2Ca5Cu7014
0.017	sacred-45	132.2	0.994	0.00	Y	Hg1Ba2Ca(173/100)Pb(17/100)Cu(143/50)O8
0.016	sacred-45	133.0	0.997	0.00	Y	Hg(9/10)Au(1/10)Ba2Ca2Cu30
0.016	sacred-45	85.0	0.994	0.00	Y	Bi2Sr(19/10)La(1/10)Ca1Cu20Y
0.014	sacred-45	128.7	0.992	0.10	Y	Hg(17/20)Re(3/20)Ba(36/25)Sr(14/25)Ca2Cu308
0.011	sacred-45	93.8	0.999	0.00	Y	Y(1/5)Eu(4/5)Ba2Cu307
0.011	3DSC	29.0	1.000	0.00	Y	Rb3C60C60
0.011	sacred-45	123.5	0.999	0.00	Y	Tl2Ba2Ca2Cd(1/10)Cu30
0.010	sacred-45	134.1	0.998	10.10	Y	Bi(8/5)Pb(2/5)Sb(1/10)Sr2Ca2Cu20
0.009	sacred-45	38.0	1.000	0.00	Y	Mg(97/100)Na(3/100)B2
0.009	sacred-45	96.0	0.999	0.00	Y	Y(4/5)Ba2Cu30(161/25)
0.006	sacred-45	111.0	0.994	0.10	Y	Bi(9/10)Nd(1/20)Tb(1/20)Pb1Sr2Ca2Cu3010
0.006	sacred-45	143.0	0.999	5.60	Y	Hg(33/50)Pb(17/50)Ba2Ca(99/50)Cu(29/10)O(42/5)
0.005	sacred-45	51.5	0.999	0.00	Y	Nd1Fe1As10(17/20)
0.005	sacred-45	36.9	1.000	0.00	Y	Mg(9/10)Li(1/10)B2
0.005	sacred-45	35.3	1.000	0.00	Y	Mg(49/50)Cr(1/50)B2
0.005	sacred-45	131.3	0.999	0.00	Y	Hg(17/20)Re(3/20)Ba(83/50)Sr(17/50)Ca2Cu308
0.004	sacred-45	106.5	0.998	1.10	Y	Tl2Ba2Ca(19/20)Y(1/20)Cu208
0.003	sacred-45	114.0	0.997	0.00	Y	Tl(1/2)Pb(1/2)Sr(4/5)Ba(1/5)Ca2Cu3U(1/25)O
0.002	sacred-45	93.0	1.000	0.00	Y	La1Ba2Cu30(137/20)
0.002	3DSC	92.3	0.970	0.60	Y	Y1Ba2Cu30Y
0.001	sacred-45	110.0	0.999	0.00	Y	Bi(8/5)Pb(2/5)Sr2Ca3Cu(97/25)Si(3/25)O12
0.001	sacred-45	20.7	1.000	0.00	Y	Nb3Al(71/100)Ge(29/100)
0.000	3DSC	19.8	1.000	0.00	Y	Nb3Al0.9B0.1
0.000	3DSC	18.9	1.000	0.00	Y	Al0.9Ge0.1Nb3
0.000	3DSC	12.2	0.999	0.00	Y	Fe1Te0.65Se0.350.05
0.000	3DSC	9.9	0.999	0.00	Y	Pr1Fe0.9Co0.1As0.7P0.301
0.000	3DSC	6.4	1.000	0.00	Y	Ca2.5Co0.5Rh4Sn13
0.000	3DSC	25.4	0.999	0.00	Y	La1Fe0.95Zn0.05As1F0.100.9
0.000	3DSC	8.8	0.999	0.00	Y	La1Fe1As0.2P0.8F0.100.9
0.000	3DSC	74.7	0.973	1.40	Y	Y1Ba2Cu2.95Zn0.050Y
0.000	3DSC	12.5	0.999	0.00	Y	Al1Nb2.1V0.9
0.000	3DSC	11.6	0.999	0.00	Y	Gd1Fe0.7Ir0.3As101
0.000	3DSC	18.1	1.000	0.00	Y	Nb3Al0.9Zn0.1
0.000	3DSC	19.8	0.995	0.00	Y	Pb2Cu0.9La1.1Cu206.19
0.000	3DSC	11.7	1.000	0.00	Y	Nb0.15Mo0.85C1
0.000	3DSC	18.3	1.000	0.00	Y	Nb75Ga7.5Al17.5
0.000	3DSC	20.0	1.000	0.00	Y	Ca0.9Pr0.1Fe1As2
0.000	3DSC	0.1	1.000	0.00	Y	Ir0.925Ru0.075
0.000	3DSC	38.7	0.999	0.00	Y	Nd1Fe0.9Ru0.1As1F0.1100.89
0.000	3DSC	29.5	0.998	0.00	Y	Nd1Fe1As0.7P0.3F0.100.9
0.000	3DSC	0.7	1.000	0.00	Y	La0.95Pr0.05Se1
0.000	3DSC	3.8	1.000	0.00	Y	Pd2Y0.7Er0.3Sn1
0.000	3DSC	18.1	1.000	0.00	Y	Nb3Sn0.9In0.1
0.000	3DSC	12.5	1.000	0.00	Y	Nb80Ga17.5Cr2.5
0.000	3DSC	56.0	0.902	3.70	Y	Y1Ba2Cu2.85Zn0.150Y
0.000	3DSC	3.3	0.999	0.00	Y	Lu1Ni0.9Cu0.1B1C1
0.000	3DSC	47.0	0.999	0.00	Y	Cu0.25Co.75Ca3Cu206.85
0.000	3DSC	24.0	0.986	0.00	Y	Cu0.75Co0.25Sr2Y1.3Ce0.7Cu208.43
0.000	3DSC	13.2	1.000	0.00	Y	Mo0.95Zr0.05C1

dist	source	T_c (K)	comp-sim	T_c err	rec	formula
0.000	3DSC	1.1	1.000	0.00	Y	Sn0.8075Ge0.1425In0.05Te1
0.000	3DSC	1.3	1.000	0.00	Y	C0.6Pd1
0.000	3DSC	53.6	0.999	0.00	Y	Hg0.2Cr0.8Sr2Cu104
0.000	3DSC	14.0	1.000	0.00	Y	Nb3Al0.7Be0.3
0.000	3DSC	0.1	1.000	0.00	Y	Ir0.925Ta0.075
0.000	3DSC	23.5	0.998	0.00	Y	Ce1Fe0.95Co0.05As1F0.100.9
0.000	3DSC	4.8	1.000	0.00	Y	Ir1.05Nb0.95
0.000	3DSC	51.4	1.000	0.00	Y	Sm0.9Th0.1Fe1As101
0.000	3DSC	3.0	1.000	0.00	Y	Nb0.85Rh1.15
0.000	3DSC	24.1	0.998	0.00	Y	La0.9Y0.1Fe0.995Mn0.005As1F0.1100.89
0.000	3DSC	34.0	1.000	0.00	Y	Ca0.9La0.1Fe1As2
0.000	3DSC	40.0	0.993	0.00	Y	Tl0.5Pb0.5Sr2Eu1.8Ce0.2Cu209
0.000	3DSC	1.6	0.999	0.00	Y	Er0.9Y0.1Pd2Sn1
0.000	3DSC	6.3	0.999	0.00	Y	La0.3Th0.7Ni1C2
0.000	3DSC	22.9	1.000	0.00	Y	Ca1Fe0.9Co0.1As1F1
0.000	3DSC	22.6	1.000	0.00	Y	Li1Hf0.7Zr0.3Ni1C11
0.000	3DSC	4.0	0.999	0.00	Y	Ce1Fe1As0.7P0.3O1
0.000	3DSC	12.8	0.999	0.00	Y	Pr0.9Sr0.1Fe1As101
0.000	3DSC	18.5	1.000	0.00	Y	Nb3Al0.9Ga0.1
0.000	3DSC	0.8	1.000	0.20	Y	Nb3Rh0.7Os0.3
0.000	3DSC	4.2	1.000	0.00	Y	Nb0.925Ru0.075
0.000	3DSC	16.0	1.000	0.00	Y	Sm1Fe0.9Co0.1As101
0.000	3DSC	36.0	1.000	0.00	Y	Ho1Fe1As1F0.100.9
0.000	3DSC	17.4	0.998	0.00	Y	Pr1Fe1As0.4P0.6F0.100.9
0.000	3DSC	9.2	1.000	0.00	Y	Ga0.3In0.7La3
0.000	3DSC	2.8	0.999	0.00	Y	Zr46.75Ti8.25Cu7.5Ni10Be27.5
0.000	3DSC	33.5	0.988	0.00	Y	Cu0.75Re0.25Sr2Y1.3Ce0.7Cu209.06
0.000	3DSC	14.1	0.999	0.00	Y	Nd1Fe0.5Ru0.5As1F0.1100.89
0.000	3DSC	46.0	0.999	0.00	Y	Nd1Fe1As1F0.300.7
0.000	3DSC	2.4	1.000	0.00	Y	La3Co3.5Ru0.5Sn13
0.000	3DSC	16.0	1.000	0.00	Y	Nb2.1Ta0.9Sn1
0.000	3DSC	46.0	1.000	0.00	Y	Ce1Fe1As1F0.300.7
0.000	3DSC	36.6	1.000	0.00	Y	Sr0.9Y0.1Cu102
0.000	3DSC	0.8	0.998	0.00	Y	Sr1.7Ca0.3V1Fe1As103
0.000	3DSC	6.2	1.000	0.00	Y	Mo5.25Nb0.75Se8
0.000	3DSC	12.0	0.995	0.00	Y	Cu0.75Ga0.25Sr2Y1.3Ce0.7Cu208.6
0.000	3DSC	28.1	0.997	0.00	Y	Pb0.6Cu0.4Sr0.8La1.2Cu105.28
0.000	3DSC	4.0	0.996	0.00	Y	Ce1Fe0.85Co0.15As1F0.100.9
0.000	3DSC	0.0	1.000	0.00	Y	Ir0.775Pt0.175Rh0.05
0.000	3DSC	35.2	0.998	0.00	Y	Pr1Fe1As0.9P0.1F0.100.9
0.000	3DSC	31.5	0.987	0.00	Y	Cu0.75W0.25Sr2Y1.3Ce0.7Cu208.81
0.000	3DSC	4.1	1.000	0.00	Y	Bi3Ni0.9Fe0.1
0.000	3DSC	25.1	1.000	0.00	Y	La1Fe1As1F0.100.9
0.000	3DSC	15.8	1.000	0.00	Y	Nd1Fe0.9Rh0.1As101
0.000	3DSC	36.3	0.999	0.00	Y	Ce1Fe1As1F0.100.9
0.000	3DSC	14.9	1.000	0.00	Y	Pr1Fe0.9Co0.1As101
0.000	3DSC	9.8	1.000	0.00	Y	Ce1Fe0.9Co0.1As1F0.100.9
0.000	3DSC	13.5	1.000	0.00	Y	Mo0.95Ti0.05C1
0.000	3DSC	19.3	0.999	0.00	Y	Ce1Fe1As0.9P0.1F0.0500.95
0.000	3DSC	12.3	1.000	0.00	Y	Nb1N1.2200.18
0.000	3DSC	1.0	0.999	0.00	Y	Zr41.2Ti13.8Cu12.5Ni10Be22.5
0.000	3DSC	2.7	1.000	0.00	Y	Nb3Rh0.9Au0.1
0.000	3DSC	2.5	1.000	0.00	Y	Mg0.9Mo5.1S6
0.000	3DSC	2.6	1.000	0.00	Y	La0.9Pr0.1Pb3
0.000	3DSC	29.2	0.999	0.00	Y	Pr1Fe1As0.8P0.2F0.100.9
0.000	3DSC	8.7	0.999	0.00	Y	Pr1Fe1As0.2P0.8F0.100.9
0.000	3DSC	3.2	0.999	0.00	Y	La1Ni0.9Cu0.1C2
0.000	3DSC	18.3	1.000	0.00	Y	Nb3Al0.9In0.1
0.000	3DSC	18.1	0.999	0.00	Y	Ce1Fe1As0.7P0.3F0.0500.95
0.000	3DSC	3.1	1.000	0.00	Y	Ru0.475V0.525
0.000	3DSC	16.0	1.000	0.00	Y	Nb3Sb0.15Sn0.7Ti0.15
0.000	3DSC	87.0	0.906	6.90	Y	Y1Ba2Cu2.97Ni0.030Y
0.000	3DSC	10.0	0.999	0.00	Y	Ce1Fe0.9Co0.1As101
0.000	3DSC	2.8	1.000	0.00	Y	Nb3Rh0.9Pt0.1
0.000	3DSC	13.5	1.000	0.00	Y	Nb0.05Mo0.95C1
0.000	3DSC	38.0	0.998	0.00	Y	La0.3Y0.7Fe1As1F0.1500.85

dist	source	T_c (K)	comp-sim	T_c err	rec	formula
0.000	3DSC	0.2	1.000	0.00	Y	Ir0.725Os0.175Rh0.1
0.000	3DSC	6.0	1.000	0.00	Y	Y0.9Th0.1Ni1C2
0.000	3DSC	82.0	0.994	0.00	Y	Y1Ba2Cu2.982Zn0.0180Y
0.000	3DSC	2.7	1.000	0.00	Y	Pb0.2375Sn0.7125In0.05Te1
0.000	3DSC	8.0	1.000	0.00	Y	Bi0.575Pb0.425
0.000	3DSC	32.0	0.998	0.00	Y	La0.8Y0.2Fe1As1F0.100.9
0.000	3DSC	0.8	0.999	0.00	Y	U1Pd2Al12.55Ga0.45
0.000	3DSC	10.0	1.000	0.00	Y	Fe0.95Co0.05Se1
0.000	3DSC	4.1	1.000	0.00	Y	Pd2Y0.9Tm0.1Sn1
0.000	3DSC	18.0	1.000	0.00	Y	Nb3Ge0.1Sn0.9
0.000	3DSC	25.2	0.994	0.00	Y	Pb0.6Cu0.4Sr0.8La1.2Cu105.29
0.000	3DSC	19.1	1.000	0.00	Y	Nb0.75Al0.175Ga0.075
0.000	3DSC	36.4	0.999	0.00	Y	Sm1Fe1As0.9P0.1F0.1200.88
0.000	3DSC	6.9	0.998	0.00	Y	Pr0.8Sr0.2Fe0.9Co0.1As101
0.000	3DSC	1.9	1.000	0.00	Y	Nb3Rh0.9Co0.1
0.000	3DSC	1.8	1.000	0.00	Y	Ba0.1Pb3Sr0.9
0.000	3DSC	12.5	0.999	0.00	Y	Sm1Fe0.7Ru0.3As1F0.1500.85
0.000	3DSC	15.8	1.000	0.00	Y	Nb3Sn0.7Al0.3
0.000	3DSC	6.2	1.000	0.00	Y	Rh0.075Zr0.925
0.000	3DSC	21.7	0.999	0.00	Y	Sm1Fe0.9Ru0.1As1F0.1500.85
0.000	3DSC	13.2	1.000	0.00	Y	Si1V2.7Zr0.3
0.000	3DSC	0.3	0.996	0.00	Y	Bi4Sr3Ca1.8Y1.2Cu40Y
0.000	3DSC	2.6	1.000	0.00	Y	Sn0.7127Pb0.2373In0.05Te1
0.000	3DSC	11.3	0.999	0.00	Y	Fe1Te0.69Se0.25S0.06
0.000	3DSC	3.5	1.000	0.00	Y	H0.6Nb0.8Pd0.2
0.000	3DSC	66.2	0.969	0.00	Y	Bi4Sr3Ca2.1Y0.9Cu40Y
0.000	3DSC	25.2	0.998	0.00	Y	Nd1Fe1As0.5P0.5F0.100.9
0.000	3DSC	11.2	1.000	0.00	Y	V3Ge0.7Al0.3
0.000	3DSC	6.4	1.000	0.00	Y	Nb5Si2.6B0.4
0.000	3DSC	21.9	0.999	0.00	Y	Pr0.65Sr0.35Fe1As0.9Sb0.101
0.000	3DSC	38.0	0.997	0.00	Y	Sr0.9La0.1Cu0.97Zn0.0302
0.000	3DSC	2.4	1.000	0.00	Y	Nb3Rh0.9Ru0.1
0.000	3DSC	26.2	0.999	0.00	Y	Sr2Mg0.5Ti0.5Fe0.95Co0.05As103
0.000	3DSC	17.8	0.999	0.00	Y	La1Fe0.7Ru0.3As1F0.1100.89
0.000	3DSC	24.6	1.000	0.00	Y	La0.9Sr0.1Fe1As101
0.000	3DSC	1.5	0.999	0.00	Y	Sn0.855Ge0.095In0.05Te1
0.000	3DSC	7.0	0.999	0.00	Y	Nd1Fe1As0.2P0.8F0.100.9
0.000	3DSC	5.7	1.000	0.00	Y	La0.9Ca0.1Fe1P101
0.000	3DSC	6.7	1.000	0.00	Y	Mo1.9Nb0.1B1C1
0.000	3DSC	14.0	1.000	0.00	Y	V3Si0.9Ge0.1
0.000	3DSC	2.9	1.000	0.00	Y	V2.7Ru0.3Si1
0.000	3DSC	8.3	1.000	0.00	Y	Nb2.1Mo0.9Al1
0.000	3DSC	4.7	1.000	0.00	Y	Ir1.15Nb0.85
0.000	3DSC	22.0	0.998	0.00	Y	Fe1.9Co0.1As2Sr4Mg1Ti106
0.000	3DSC	10.8	1.000	0.00	Y	V0.75Ge0.175Al0.075
0.000	3DSC	90.2	0.987	0.00	Y	Y1Ba2Cu2.7Ag0.30Y
0.000	3DSC	14.1	1.000	0.00	Y	Al1Cr0.3Nb2.7
0.000	3DSC	7.5	0.998	0.00	Y	Pr1Fe0.9Co0.1As0.5P0.501
0.000	3DSC	9.5	1.000	0.00	Y	Nb3Rh0.3Au0.7
0.000	3DSC	41.5	0.999	0.00	Y	Sr0.9Pr0.1Cu102
0.000	3DSC	9.0	1.000	0.00	Y	C1Hf0.15Mo0.85
0.000	3DSC	87.6	0.981	0.00	Y	Y1Ba2Cu2.55Ag0.450Y
0.000	3DSC	12.8	1.000	0.00	Y	Nb0.75Au0.175Pt0.075
0.000	3DSC	16.5	0.999	0.00	Y	Nb0.95Mg0.05N0.9200.08
0.000	3DSC	50.5	1.000	0.00	Y	Tb1Fe1As1F0.100.9
0.000	3DSC	62.3	0.998	0.00	Y	Hg0.9Cr0.1Sr2Cu104
0.000	3DSC	1.7	1.000	0.00	Y	Sn0.9215Ge0.0285In0.05Te1
0.000	3DSC	31.4	0.998	0.00	Y	La0.5Y0.5Fe1As1F0.100.9
0.000	3DSC	35.7	0.999	0.00	Y	Ca0.7La0.3Fe1As1H1
0.000	3DSC	12.7	1.000	0.00	Y	Nb0.85Mo0.15C1
0.000	3DSC	1.6	0.999	0.00	Y	Sn0.9405Ge0.0095In0.05Te1
0.000	3DSC	25.8	0.999	0.00	Y	Sr1.9Ca0.1V1Fe1As103
0.000	3DSC	16.1	1.000	0.00	Y	Nb3Sn0.1Al0.9
0.000	3DSC	28.2	1.000	0.00	Y	Ca0.9La0.1Fe1As1H1
0.000	3DSC	21.3	0.999	0.00	Y	Nd1Fe0.7Ru0.3As1F0.1100.89
0.000	3DSC	14.0	0.999	0.00	Y	Bi1Pb1Sr0.3Pr0.3Cu106

dist	source	T_c (K)	comp-sim	T_c err	rec	formula
0.000	3DSC	13.7	1.000	0.00	Y	Nb0.85Zr0.15N1
0.000	3DSC	18.2	1.000	0.00	Y	Nb3Sn0.9Ga0.1
0.000	3DSC	24.5	0.988	0.00	Y	Cu0.75V0.25Sr2Y1.3Ce0.7Cu208.77
0.000	3DSC	11.5	1.000	0.00	Y	Nb3Au0.7Pt0.3
0.000	3DSC	8.2	1.000	0.00	Y	Fe0.95Co0.05Se0.3Te0.7
0.000	3DSC	45.9	1.000	0.00	Y	Gd1Fe1As1H0.300.7
0.000	3DSC	12.7	1.000	0.00	Y	Mo0.95Hf0.05C1
0.000	3DSC	7.5	1.000	0.00	Y	Lu0.9Ho0.1Ru1B2
0.000	3DSC	10.6	1.000	0.00	Y	V3Si0.1Ga0.9
0.000	3DSC	10.0	0.995	0.00	Y	Cu0.75Ce0.25Sr2Y1.3Ce0.7Cu208.91
0.000	3DSC	18.1	0.999	0.00	Y	Bi0.15Nb3Sn0.7Ti0.15
0.000	3DSC	14.8	1.000	0.00	Y	P0.3Si0.7V3
0.000	3DSC	42.0	0.999	1.40	Y	Sr0.9La0.1Cu0.99Zn0.0102
0.000	3DSC	23.4	0.998	0.00	Y	Pr1Fe1As0.6P0.4F0.100.9
0.000	3DSC	100.0	0.994	0.00	Y	Cu0.5S0.5Sr2Ca2Cu3012
0.000	3DSC	19.9	0.999	0.00	Y	La1Fe0.9Zn0.1As1F0.100.9
0.000	3DSC	13.2	1.000	0.00	Y	V2.7Zr0.3Si1
0.000	3DSC	6.4	1.000	0.00	Y	Li2Pd2.85Pt0.15B1
0.000	3DSC	3.5	1.000	0.00	Y	Pb0.35Sn0.53In0.12Te1
0.000	3DSC	7.8	1.000	0.00	Y	Nb5Si2.4B0.6
0.000	3DSC	34.9	0.997	0.00	Y	Pb0.5Cu0.5Sr0.8La1.2Cu105.1
0.000	3DSC	18.0	1.000	0.00	Y	Nb3Sn0.9Ge0.1
0.000	3DSC	28.0	0.998	0.00	Y	Nd1Fe1As0.6P0.4F0.100.9
0.000	3DSC	4.5	1.000	0.00	Y	La0.9Th0.1Ni1C2
0.000	3DSC	3.5	0.999	0.00	Y	Pd2Y0.9dy0.1Sn1
0.000	3DSC	30.0	0.998	0.00	Y	Sr2V0.9Mg0.1Fe1As103
0.000	3DSC	98.0	1.000	0.00	Y	Cu1Cr0.9Fe0.102
0.000	3DSC	31.3	0.994	0.00	Y	Pb2Cu0.9La1.1Cu206.24
0.000	3DSC	4.6	0.999	0.00	Y	Mo1.7Ta0.3B1C1
0.000	3DSC	82.8	0.987	0.00	Y	Y1Ba2Cu2.975Zn0.0250Y
0.000	3DSC	12.8	1.000	0.00	Y	Si1V2.7Nb0.3
0.000	3DSC	11.1	1.000	0.00	Y	V0.75Si0.175Al0.075
0.000	3DSC	21.2	0.998	0.00	Y	La1Fe1As0.5P0.5F0.0500.95
0.000	3DSC	1.7	1.000	0.00	Y	Sn0.9025Ge0.0475In0.05Te1
0.000	3DSC	3.9	1.000	0.00	Y	Pd2Y0.9Er0.1Sn1
0.000	3DSC	12.8	1.000	0.00	Y	V2.7Nb0.3Si1
0.000	3DSC	30.0	0.999	0.00	Y	Sr0.9La0.1Cu0.99Ni0.0102
0.000	3DSC	16.1	1.000	0.00	Y	Nb3Ga0.9Sn0.1
0.000	3DSC	22.0	0.998	0.00	Y	Pr0.6Sr0.4Fe1As0.9Sb0.101
0.000	3DSC	18.2	1.000	0.00	Y	Nb3Ti0.1Sn0.9
0.000	3DSC	32.6	0.999	0.00	Y	Sr0.7Y0.3Cu102
0.000	3DSC	6.8	1.000	0.00	Y	Nb3Sn0.3Sb0.7
0.000	3DSC	7.2	1.000	0.00	Y	La0.7Th0.3Ni1C2
0.000	3DSC	2.9	0.999	0.00	Y	Nd1Fe1P1F0.100.9
0.000	3DSC	50.5	1.000	0.00	Y	Nd1Fe1As1F0.100.9
0.000	3DSC	45.0	1.000	0.00	Y	Dy1Fe1As1F0.100.9
0.000	3DSC	57.0	0.996	0.00	Y	Cu0.5C0.5Ca3Cu206.6
0.000	3DSC	27.0	0.998	0.00	Y	La1Fe1As0.6P0.4F0.100.9
0.000	3DSC	10.0	1.000	0.00	Y	Al0.175Nb0.775Zr0.05
0.000	3DSC	7.2	1.000	0.00	Y	Pb0.95Ti0.025Bi0.025
0.000	3DSC	17.6	1.000	0.00	Y	Nb3Al0.9Se0.1
0.000	3DSC	23.6	0.998	0.00	Y	La1Fe1As0.9P0.1F0.0500.95
0.000	3DSC	0.1	1.000	0.00	Y	Ag0.825Sn0.175
0.000	3DSC	11.0	1.000	0.00	Y	La1Fe0.9Co0.1As101
0.000	3DSC	17.5	0.988	0.00	Y	Cu0.75Ge0.25Sr2Y1.3Ce0.7Cu208.59
0.000	3DSC	4.4	1.000	0.00	Y	Pd2Y0.9Yb0.1Sn1
0.000	3DSC	91.3	0.965	0.20	Y	Y1Ba2Cu2.85Ag0.150Y
0.000	3DSC	3.7	0.999	0.00	Y	La0.9Sr0.1Ni1As101
0.000	3DSC	5.1	1.000	0.00	Y	Nb3Rh0.7Pt0.3
0.000	3DSC	0.9	1.000	0.00	Y	Os0.075W0.925
0.000	3DSC	4.8	0.999	0.00	Y	Lu0.9Tb0.1Ru1B2
0.000	3DSC	15.8	1.000	0.00	Y	Nb3Sn0.7Sb0.3
0.000	3DSC	8.8	1.000	0.00	Y	V3Ge0.9Al0.1
0.000	3DSC	38.7	0.996	0.00	Y	Pb0.5Cu0.5Sr0.8La1.2Cu105.01
0.000	3DSC	2.3	1.000	0.00	Y	La0.9Y0.1Ni1C2
0.000	3DSC	8.6	1.000	0.00	Y	V0.75Ge0.225Al0.025

dist	source	T_c (K)	comp-sim	T_c err	rec	formula
0.000	3DSC	19.8	1.000	0.00	Y	Nb3Al0.9Be0.1
0.000	3DSC	10.9	0.999	0.00	Y	La1Fe0.9Ir0.1As101
0.000	3DSC	2.5	1.000	0.00	Y	Nb3Rh0.9Pd0.1
0.000	3DSC	9.0	0.999	0.00	Y	Nd1Fe0.9Co0.1As1F0.1100.89
0.000	3DSC	5.2	0.998	0.00	Y	Y0.45Dy0.55Ni2B2C1
0.000	3DSC	0.1	1.000	0.00	Y	Ir0.46Re0.115Rh0.425
0.000	3DSC	14.0	1.000	0.00	Y	Si0.9V3Ge0.1
0.000	3DSC	14.7	0.998	0.00	Y	Pr1Fe0.9Co0.1As0.9P0.101
0.000	3DSC	1.8	1.000	0.00	Y	Sn0.9025Pb0.0475In0.05Te1
0.000	3DSC	36.2	0.999	0.00	Y	La0.7Y0.3Fe1As1F0.1500.85
0.000	3DSC	28.6	0.999	0.00	Y	La1Fe1As0.9Sb0.1F0.200.8
0.000	3DSC	13.7	0.999	0.00	Y	Sr1.8Ca0.2V1Fe1As103
0.000	3DSC	9.1	1.000	0.00	Y	Nb3Al0.3Ge0.7
0.000	3DSC	14.0	1.000	0.00	Y	La1Fe0.9Rh0.1As101
0.000	3DSC	43.5	1.000	0.00	Y	Pr1Fe1As1F0.100.9
0.000	3DSC	17.7	1.000	0.00	Y	Nb2.1Zr0.9Sn1
0.000	3DSC	6.4	0.999	0.00	Y	Lu0.9Gd0.1Ru1B2
0.000	3DSC	28.0	0.999	0.00	Y	Fe2As2Sr4Mg0.4Ti1.606
0.000	3DSC	18.0	0.998	0.00	Y	As0.15Nb3Sn0.7Tl0.15
0.000	3DSC	12.1	1.000	0.00	Y	Nb0.95Mo0.05C1
0.000	3DSC	1.6	0.999	0.00	Y	Sn0.6175Pb0.3325In0.05Te1
0.000	3DSC	10.7	1.000	0.00	Y	V3Ga0.9Sn0.1
0.000	3DSC	47.0	0.999	0.00	Y	Er1Sr2Cu2.85Re0.150Y
0.000	3DSC	2.8	1.000	0.00	Y	Sn0.8075Pb0.1425In0.05Te1
0.000	3DSC	11.8	1.000	0.00	Y	Ta0.05Mo0.95C1
0.000	3DSC	14.2	0.995	0.00	Y	Pr1Fe0.9Co0.1As0.95P0.0501
0.000	3DSC	38.0	1.000	0.20	Y	La1Fe1As1H0.300.7
0.000	3DSC	1.6	0.999	0.00	Y	Pb0.3325Sn0.6175In0.05Te1
0.000	3DSC	81.7	0.992	0.00	Y	Bi4Sr3Ca2.7Y0.3Cu40Y
0.000	3DSC	2.1	1.000	0.00	Y	Pd2Zr0.9Nb0.1Al1
0.000	sacred-45	117.0	0.997	0.00	Y	C(1/2)Ba2Ca3Cu(9/2)O11
0.000	sacred-45	20.4	1.000	0.00	Y	Nb3Al(19/25)Ge(6/25)
0.000	sacred-45	110.2	0.995	0.10	Y	Bi(17/10)Pb(3/10)Sr2Ca2Cu30
0.000	sacred-45	99.0	0.999	0.00	Y	Y1Ba(39/20)La(1/20)Cu30(69/10)
0.000	3DSC	85.8	0.989	0.70	Y	Bi4Sr3Ca2.55Y0.45Cu40Y
0.000	3DSC	12.5	1.000	0.00	Y	Au0.7Nb3Pt0.3
0.000	3DSC	1.7	1.000	0.00	Y	Ru2Sc1.2
0.000	3DSC	17.0	1.000	0.00	Y	Nb3Sb0.1Sn0.9
0.000	3DSC	10.0	1.000	0.00	Y	Fe0.95Ni0.05Se1
0.000	3DSC	28.0	0.993	0.00	Y	Pb0.5Cu0.5Sr0.8La1.2Cu105.16
0.000	3DSC	1.3	1.000	0.00	Y	Lu0.275Rh0.725
0.000	3DSC	2.3	0.999	0.00	Y	Sn0.6646Pb0.2854In0.05Te1
0.000	3DSC	10.9	0.999	0.00	Y	Pr1Fe0.9Co0.1As0.8P0.201
0.000	3DSC	20.9	0.998	0.00	Y	La1Fe1P0.6As0.4F0.100.9
0.000	3DSC	43.5	1.000	0.00	Y	Sr0.9Gd0.1Cu102
0.000	3DSC	41.0	0.996	0.00	Y	Cu0.5C0.5Sr2Ca1Cu5010.8
0.000	3DSC	49.0	1.000	0.00	Y	Ca0.875Pr0.125Fe2As2
0.000	3DSC	23.6	0.998	0.00	Y	La1Fe0.9Ru0.1As1F0.1100.89
0.000	3DSC	11.2	1.000	0.00	Y	V0.75Ge0.025Al0.225
0.000	3DSC	6.6	0.999	0.00	Y	Li2Pd2.55Ni0.45B1
0.000	3DSC	9.6	1.000	0.00	Y	Cd0.1In0.9La3
0.000	3DSC	3.9	0.999	0.00	Y	Sn0.8In0.2Te0.85Se0.15
0.000	3DSC	10.4	1.000	0.00	Y	Nb48.15Zr42.5Ta9.35
0.000	3DSC	7.4	0.999	0.00	Y	Y0.7Th0.3Ni1C2
0.000	3DSC	9.6	1.000	0.00	Y	Sm1Fe0.9Rh0.1As101
0.000	3DSC	9.7	1.000	0.00	Y	La1Fe1As0.7P0.301
0.000	3DSC	6.2	0.999	0.00	Y	Lu2Ir3Si4.75Ge0.25
0.000	3DSC	0.1	1.000	0.00	Y	Cu0.825Ge0.175
0.000	3DSC	10.8	1.000	0.00	Y	Nb3Rh0.1Au0.9
0.000	3DSC	1.6	1.000	0.00	Y	Sn0.874Ge0.076In0.05Te1
0.000	3DSC	9.6	1.000	0.00	Y	La3In0.9Cd0.1
0.000	3DSC	0.4	1.000	0.00	Y	La0.7Pr0.3Pb3
0.000	3DSC	13.9	1.000	0.00	Y	Al0.3Ga0.7V3
0.000	3DSC	47.6	0.901	2.50	Y	Y1Ba2Cu2.7Zn0.30Y
0.000	3DSC	9.4	1.000	0.00	Y	La3In0.7Al0.3
0.000	3DSC	85.0	0.978	0.10	Y	Y1Ba2Cu2.988Zn0.0120Y

dist	source	T_c (K)	comp-sim	T_c err	rec	formula
0.000	3DSC	18.2	0.999	0.00	Y	Sm1Fe0.9Zn0.1As1F0.200.8
0.000	3DSC	16.1	1.000	0.00	Y	Nb3Ga0.9Zn0.1
0.000	3DSC	7.3	1.000	0.00	Y	Lu0.9Dy0.1Ru1B2
0.000	3DSC	64.9	0.904	9.30	Y	Y1Ba2Cu2.925Zn0.0750Y
0.000	3DSC	24.0	0.997	0.00	Y	Fe1.8Co0.2As2Sr4Mg1Ti106
0.000	3DSC	3.3	0.999	0.00	Y	Sn0.8In0.2Te0.95Se0.05
0.000	3DSC	5.5	0.998	0.00	Y	Nd0.6Sm0.4F0.3Bi1S200.5
0.000	3DSC	33.4	0.986	0.00	Y	Pb2Cu0.9La1.1Cu206.28
0.000	3DSC	3.8	1.000	0.00	Y	V3Ga0.1Sn0.9
0.000	3DSC	12.5	0.997	0.00	Y	Fe2As2Ca4Al1.34Ti0.6606
0.000	3DSC	25.5	1.000	0.00	Y	La1Fe0.98Zn0.02As1F0.100.9
0.000	3DSC	2.1	1.000	0.00	Y	Sn0.85Ag0.15Te1
0.000	3DSC	34.1	0.998	0.00	Y	Nd1Fe1As0.8P0.2F0.100.9
0.000	3DSC	2.3	0.999	0.00	Y	Pr1Fe1P1F0.100.9
0.000	3DSC	18.1	1.000	0.00	Y	Ga0.15Nb3Sn0.7Ti0.15
0.000	3DSC	16.0	0.996	0.00	Y	Fe1.7Co0.3As2Sr4Mg1Ti106
0.000	3DSC	2.7	1.000	0.00	Y	Nb0.9Mo2.1Al1
0.000	3DSC	3.9	1.000	0.00	Y	La1Ni1As1F0.100.9
0.000	3DSC	4.6	0.999	0.00	Y	Lu2Ir3Si4.25Ge0.75
0.000	3DSC	4.0	1.000	0.00	Y	Pr1Fe0.7Co0.3As101
0.000	3DSC	17.9	1.000	0.00	Y	Nb0.75Al0.225Ge0.025
0.000	3DSC	13.5	0.999	0.00	Y	Fe0.95Zn0.05Se0.3Te0.7
0.000	3DSC	4.5	0.998	0.00	Y	Pr0.8Sr0.2Fe0.7Co0.3As101
0.000	3DSC	17.8	0.999	0.00	Y	Co.25Nb0.75Nb0.85Ti0.15
0.000	3DSC	44.9	0.999	0.00	Y	Ce1Fe1As1H0.300.7
0.000	3DSC	2.3	1.000	0.00	Y	Nb3Rh0.9Os0.1
0.000	3DSC	0.2	1.000	0.00	Y	Ir0.925Nb0.075
0.000	3DSC	6.0	1.000	0.00	Y	La1Fe1As0.1P0.901
0.000	3DSC	4.6	1.000	0.00	Y	Nb3Rh0.7Au0.3
0.000	3DSC	2.3	0.998	0.00	Y	Y2Pd1Si2.25Co.75
0.000	3DSC	24.0	1.000	0.00	Y	Mg1B0.85Co.15
0.000	3DSC	0.0	1.000	0.00	Y	Ir0.83Pd0.045Pt0.125
0.000	3DSC	45.0	1.000	0.10	Y	Tb0.9Th0.1Fe1As101
0.000	3DSC	7.9	1.000	0.00	Y	Nb3Rh0.1Pt0.9
0.000	3DSC	9.2	1.000	0.00	Y	La3In0.7Ga0.3
0.000	3DSC	7.4	1.000	0.00	Y	Nb3Rh0.3Pt0.7
0.000	3DSC	5.6	0.999	0.00	Y	Sm1Fe0.9Ni0.1As1F0.1200.88
0.000	3DSC	25.6	0.985	0.00	Y	Bi4Sr3Ca1.95Y1.05Cu40Y
0.000	sacred-45	100.1	1.000	0.10	Y	Y3Ba5Cu8018
0.000	3DSC	2.3	0.999	0.00	Y	Ni2Zr0.7Nb0.3Ga1
0.000	3DSC	34.8	0.998	0.00	Y	La0.7Y0.3Fe1As1F0.100.9
0.000	3DSC	15.2	1.000	0.00	Y	Nb77.5Sn5.6Al11.3Ge5.6
0.000	3DSC	34.8	0.995	0.00	Y	Pb0.6Cu0.4Sr0.8La1.2Cu105.11
0.000	3DSC	7.0	0.997	0.00	Y	Ge0.3Cu0.7Sr2Ca3Cu4011.05
0.000	3DSC	12.2	0.998	0.00	Y	Pr1Fe0.9Co0.1As0.85P0.1501
0.000	3DSC	57.6	0.905	4.90	Y	Y1Ba2Cu2.9Zn0.10Y
0.000	3DSC	4.1	0.999	0.00	Y	Pd2Y0.9Ho0.1Sn1
0.000	3DSC	44.0	1.000	0.00	Y	Sr0.9Sm0.1Cu102
0.000	3DSC	28.7	0.989	0.00	Y	Pb0.6Cu0.4Sr0.8La1.2Cu105.06
0.000	3DSC	56.5	0.973	0.40	Y	Y1Ba2Cu2.8Zn0.20Y
0.000	3DSC	8.9	1.000	0.00	Y	C1Mo0.85Ta0.15
0.000	3DSC	25.2	0.995	0.00	Y	Pb0.5Cu0.5Sr0.8La1.2Cu105.25
0.000	3DSC	18.1	1.000	0.00	Y	Al0.9Nb3Sb0.1
0.000	3DSC	9.8	1.000	0.00	Y	V0.75Si0.025Al0.225
0.000	3DSC	7.0	0.999	0.00	Y	La0.9Y0.1Fe1As1F0.100.9
0.000	3DSC	18.1	1.000	0.00	Y	Nb3Sn0.9Al0.1
0.000	3DSC	41.5	0.915	9.50	Y	Sr0.9La0.1Cu102
0.000	3DSC	107.0	0.999	0.00	Y	V0.5Sr2.7Ca2.3Cu4.5011.6
0.000	3DSC	21.0	0.989	0.00	Y	Cu0.75Ti0.25Sr2Y1.3Ce0.7Cu208.71
0.000	3DSC	19.1	1.000	0.00	Y	Nb0.75Al0.225Ga0.025
0.000	3DSC	13.9	1.000	0.00	Y	Mo0.95V0.05C1
0.000	3DSC	15.4	1.000	0.00	Y	Ca1Fe0.9Ni0.1As1F1
0.000	3DSC	79.7	0.980	0.00	Y	Bi4Sr3Ca2.4Y0.6Cu40Y
0.000	3DSC	2.9	1.000	0.00	Y	Lu1Ni0.9V0.1B1C1
0.000	3DSC	1.2	1.000	0.00	Y	N0.6Ti1
0.000	3DSC	2.1	1.000	0.00	Y	Re0.075Ta0.925

dist	source	T_c (K)	comp-sim	T_c err	rec	formula
0.000	3DSC	16.4	1.000	0.00	Y	Nb0.75Al0.225Si0.025
0.000	3DSC	2.1	0.999	0.00	Y	Sn0.8549Pb0.0951In0.05Te1
0.000	3DSC	17.7	1.000	0.00	Y	Al0.9Ga0.1Nb3
0.000	3DSC	82.0	0.984	0.00	Y	Y0.4Cu0.6Ba1.3La0.7Cu307
0.000	3DSC	4.0	1.000	0.00	Y	Pd2Y0.7Yb0.3Sn1
0.000	3DSC	26.0	0.997	0.00	Y	La1Fe1As0.8P0.2F0.100.9
0.000	3DSC	22.6	0.999	0.00	Y	Nd1Fe1As0.4P0.6F0.100.9
0.000	3DSC	19.9	0.998	0.00	Y	La1Fe0.85Zn0.15As1F0.100.9
0.000	3DSC	15.8	1.000	0.00	Y	La1Fe1As0.7P0.3F0.0500.95
0.000	3DSC	7.2	1.000	0.00	Y	V3Ga0.7Sn0.3
0.000	3DSC	90.0	0.997	0.00	Y	Cu0.5C0.5Sr2Ca1Cu207
0.000	3DSC	21.3	0.999	0.00	Y	La1Fe1As0.4P0.6F0.100.9
0.000	3DSC	8.8	1.000	0.00	Y	Lu0.9Er0.1Ru1B2
0.000	3DSC	8.7	0.999	0.00	Y	Lu0.9Tm0.1Ru1B2
0.000	3DSC	81.0	0.984	0.00	Y	Y0.4Cu0.6Ba1.4La0.6Cu306.95
0.000	3DSC	14.4	1.000	0.00	Y	Ge0.1Si0.9V3
0.000	3DSC	8.0	0.987	0.00	Y	Cu0.75Cr0.25Sr2Y1.3Ce0.7Cu208.87
0.000	3DSC	2.9	1.000	0.00	Y	Si1V2.7Ru0.3
0.000	3DSC	5.9	1.000	0.00	Y	Nb5Si2.8B0.2
0.000	3DSC	55.6	0.999	0.00	Y	Gd1Fe1As1H0.100.9
0.000	3DSC	3.8	1.000	0.00	Y	In0.075Sn0.925
0.000	3DSC	13.6	1.000	0.00	Y	La1Fe0.5Ru0.5As1F0.1100.89
0.000	3DSC	89.8	0.977	0.10	Y	Bi4Sr3Ca2.25Y0.75Cu40Y

7 Sacred-45 subset: per-family recovery

The 45 sacred targets (five per family across nine families, from a seed-42 test split, never presented to the model in any form) carry clean per-family labels; the full per-target rows are in the table above (source **sacred-45**). Per-family recovery under the joint criterion:

Family	n	comp-sim	T_c err (K)	recovered
MgB2	5	1.000	0.0	5/5
YBCO	5	0.999	0.2	5/5
Iron-based	5	0.998	1.4	5/5
Conventional	5	0.997	0.1	5/5
Hg-cuprate	5	0.996	1.1	5/5
Other	5	0.995	2.2	5/5
Bi-cuprate	5	0.994	4.6	5/5
LSCO	5	0.988	8.2	4/5
Tl-cuprate	5	0.986	1.4	5/5